

DSA 0604

Data Handling and Visualization

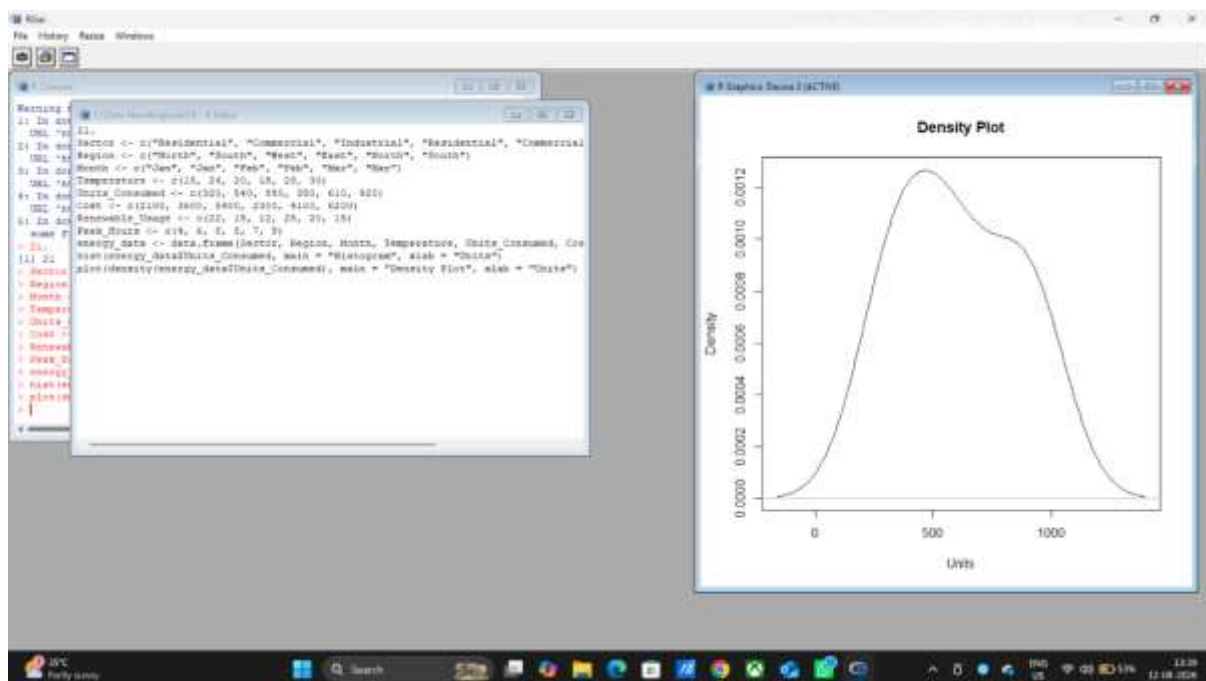
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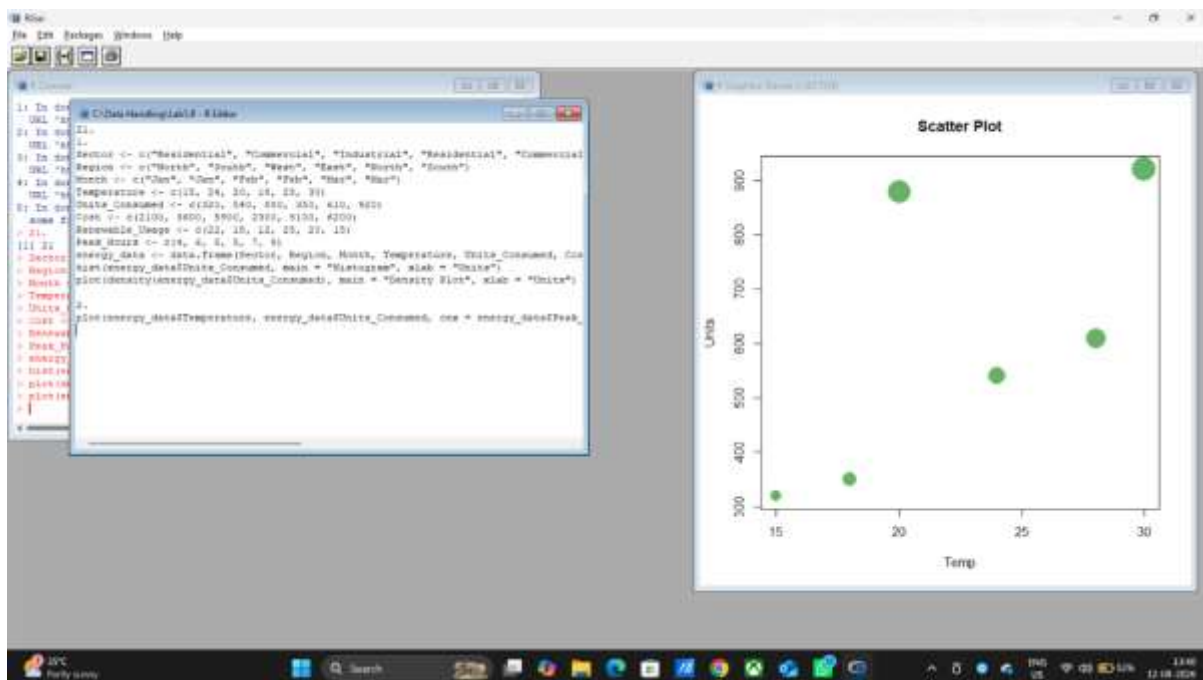
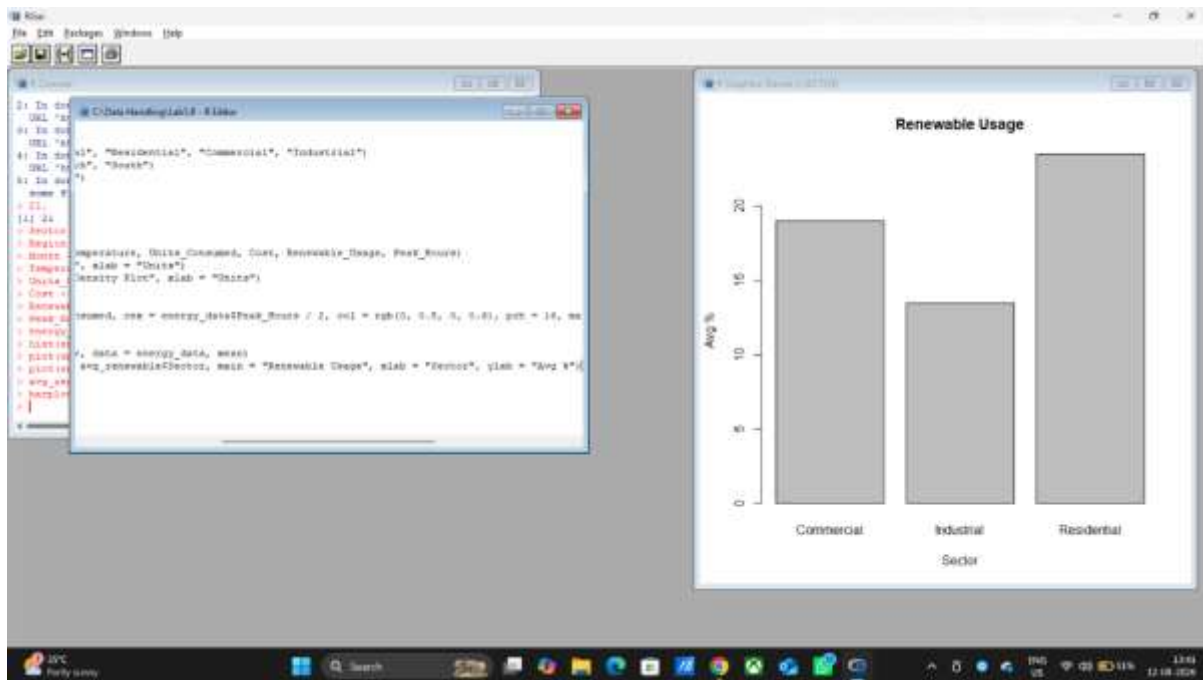
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Set 21

Dataset: Energy_Consumption_Data.csv

1. Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.
2. Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.
3. Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.
4. Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard



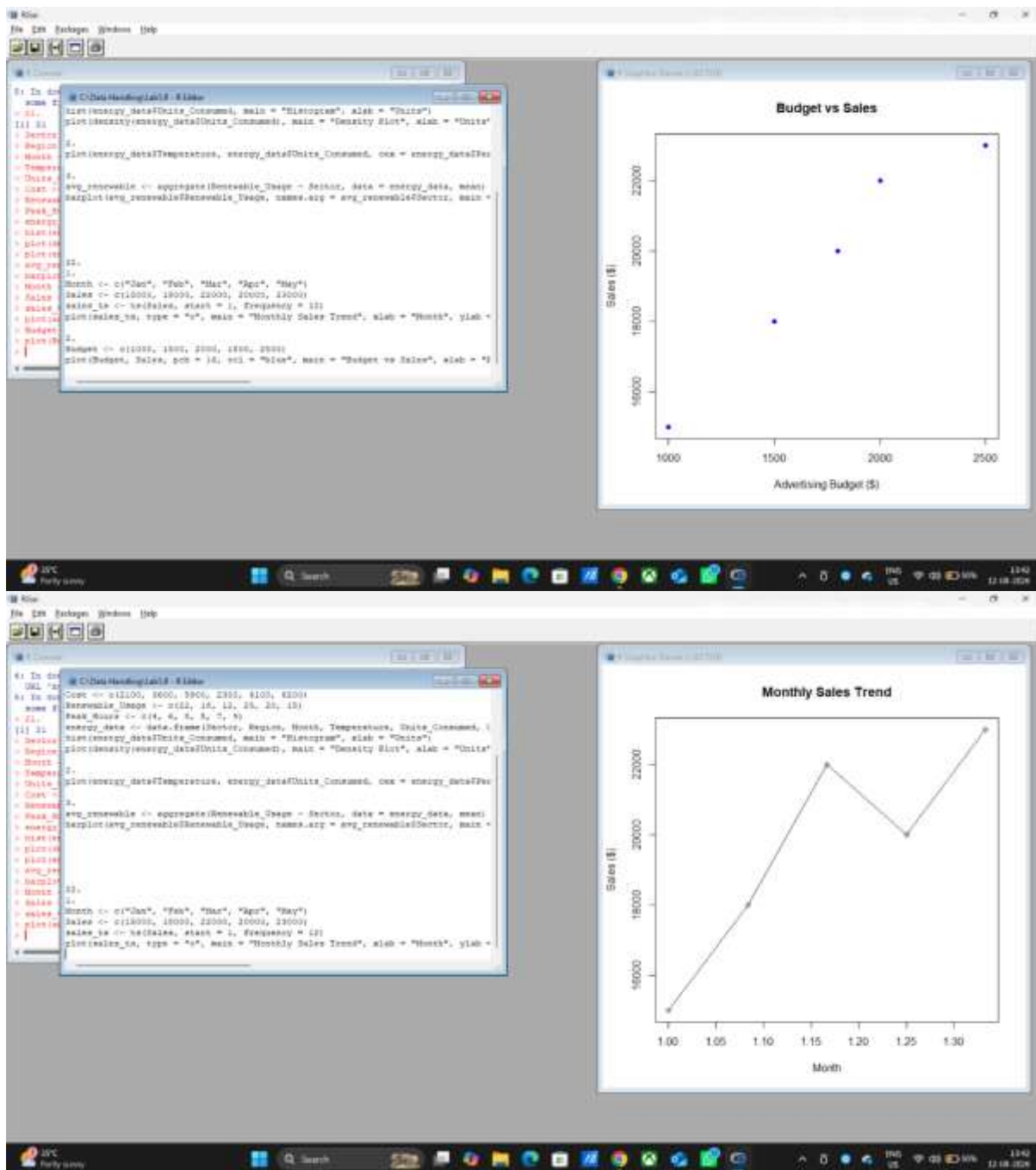


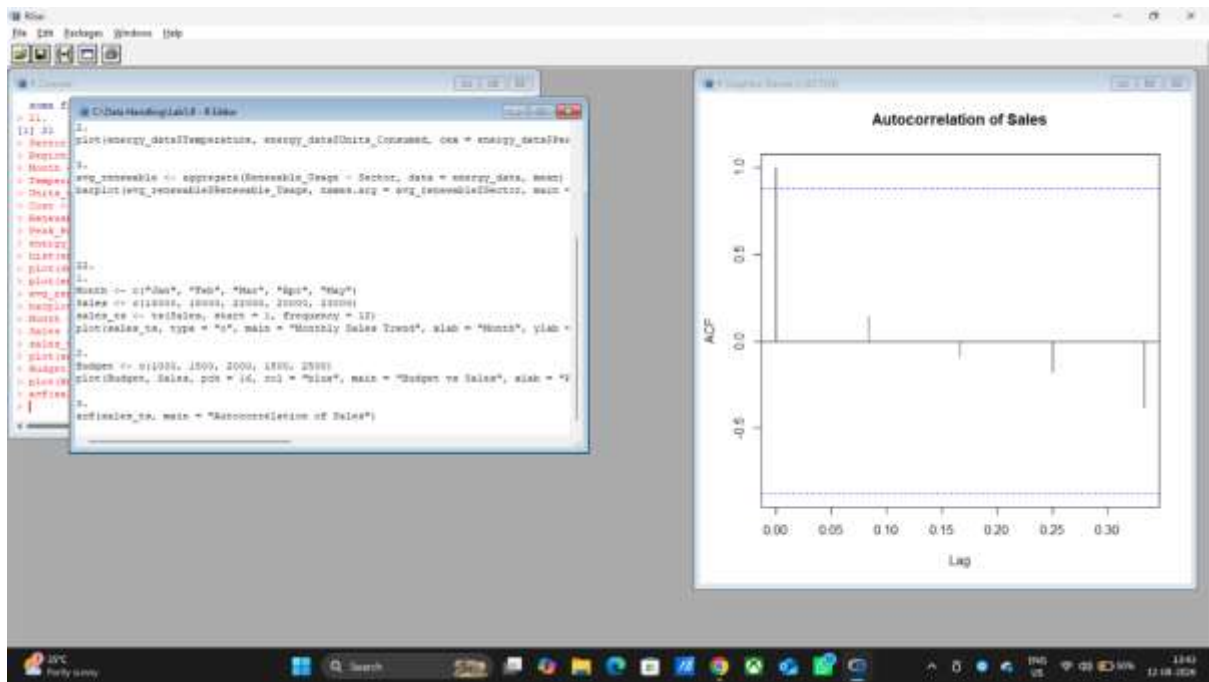
Set 22: Time Series Analysis

Dataset: Monthly Sales Data

1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.
2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.
3. In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.

4. In R, create an autocorrelation plot to identify seasonality in the time series data.

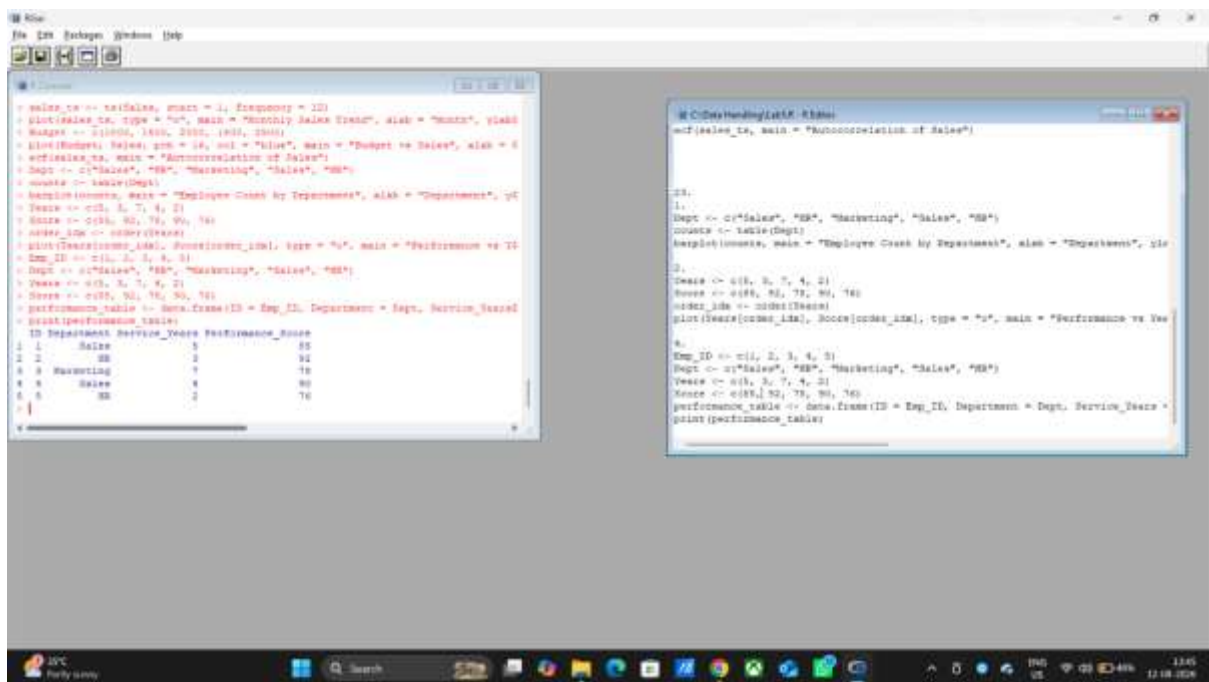
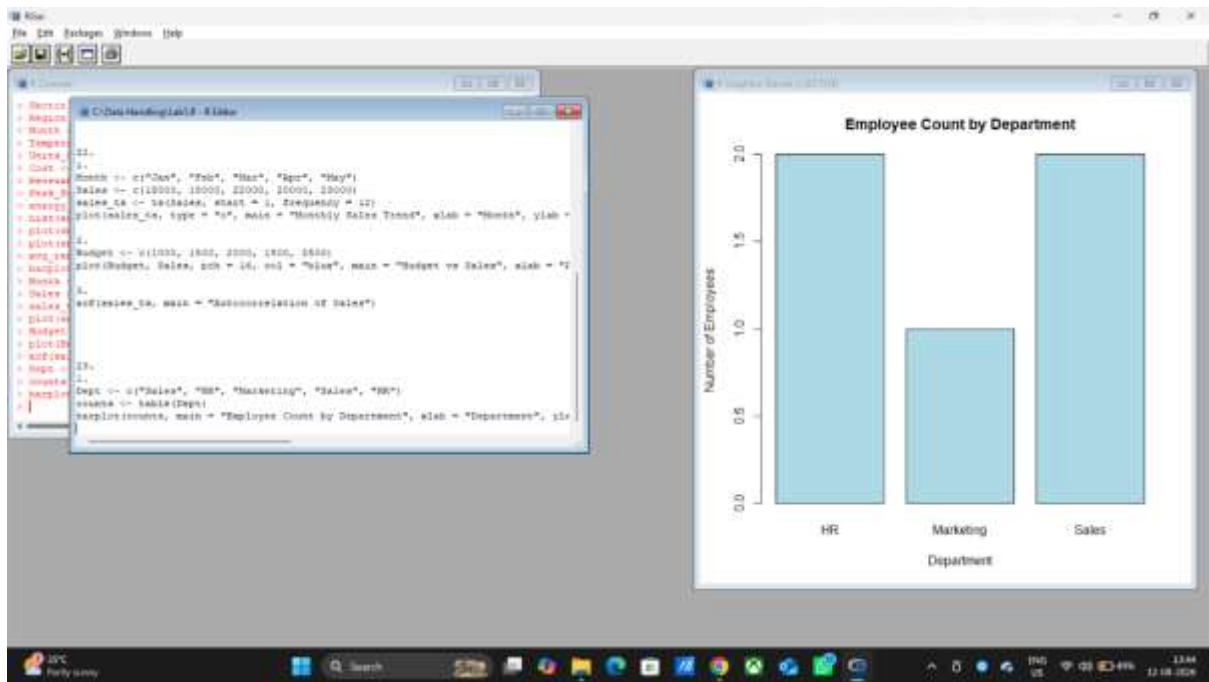


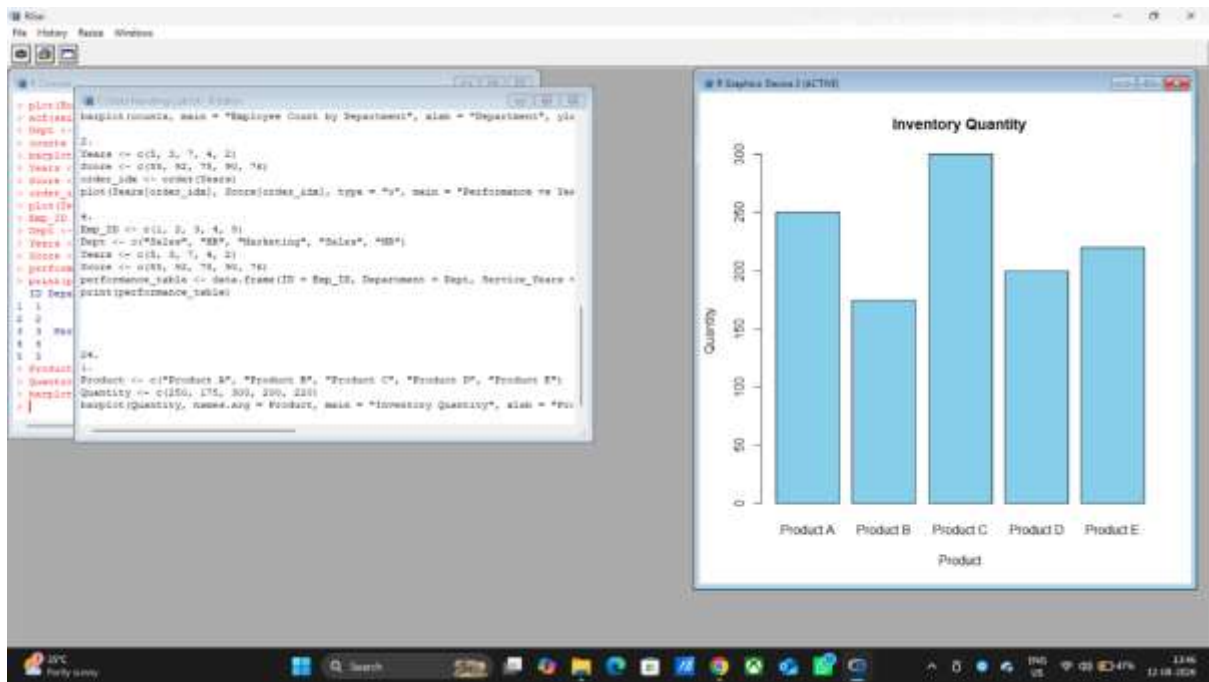


Set 23: Employee Performance Analysis

Dataset: Employee Performance

1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.
2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
3. In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.
4. In R, develop a table showing the performance data for each employee. Label the table elements.

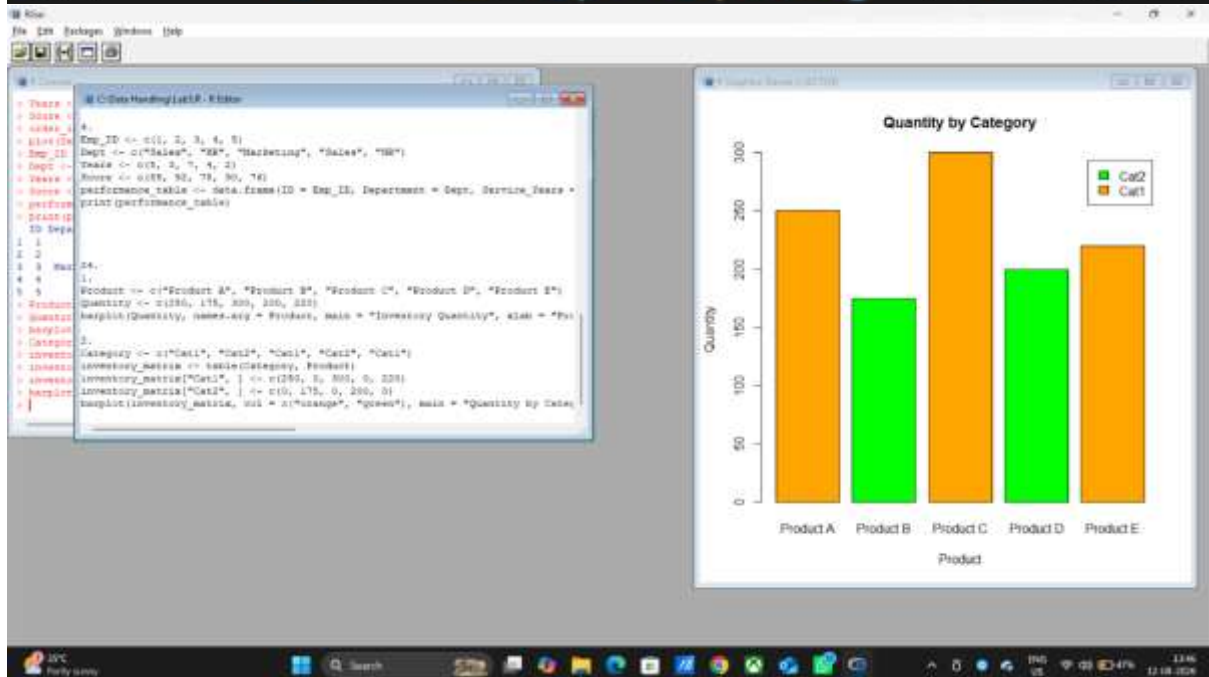
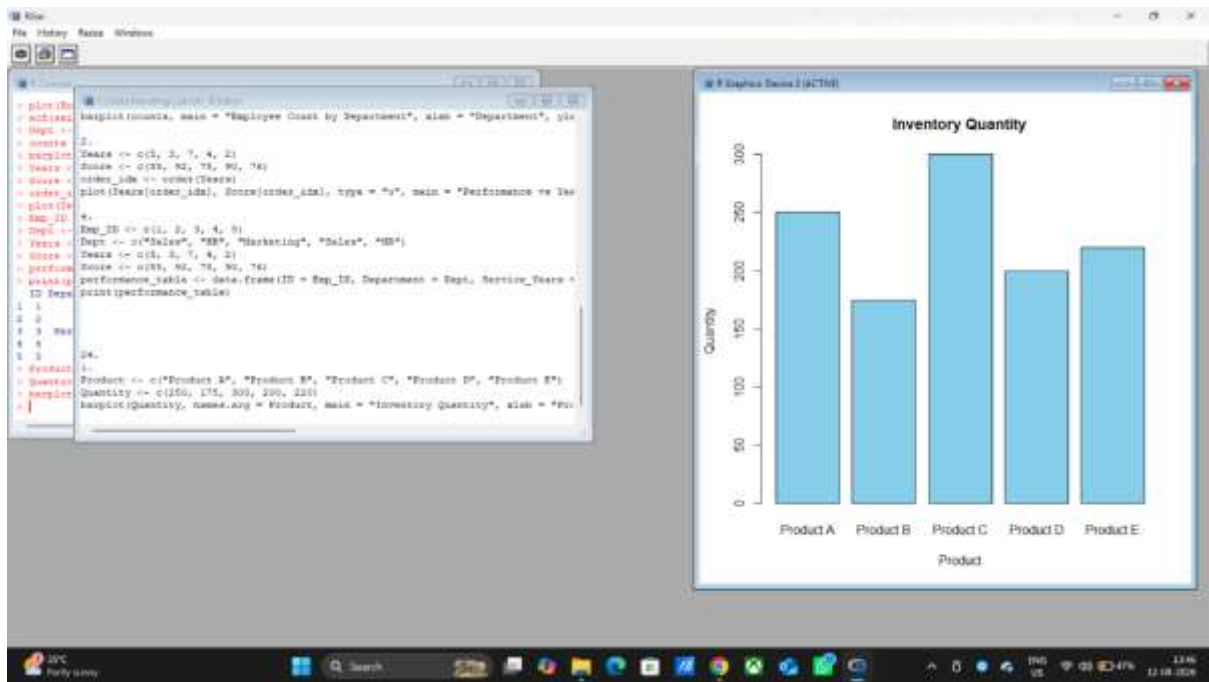


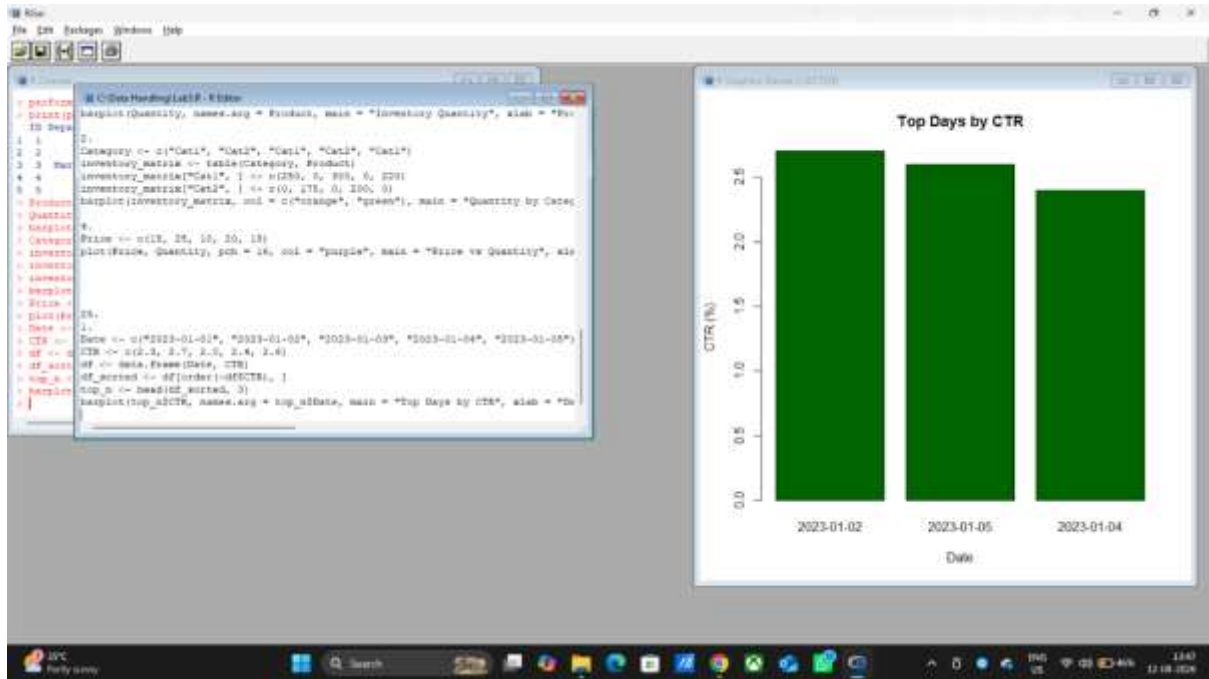


Set 24 Product Inventory Management

Dataset: Product Inventory

1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.
3. In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.
4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights





Set 25: Website Traffic Analysis

Dataset: Website Traffic

1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.
2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.
3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.
4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source..

